

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1803

Loudon 345 kV
September 20, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1803. The Interconnection Customer proposes a 151.5 MW wind facility interconnecting at the Loudon 345 kV substation in the Virginia area. The request was received on July 01, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 584 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$95,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Nauvoo - City W Diesel Plant 345.0 kV L, the Nauvoo - Roberts Sw Sta 345.0 kV loads to 102.3% of its 2,021 MVA rating. Project DFAX: 6.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Nauvoo - Roberts Sw Sta 345.0 kV	Reconductor line	\$7,400,000	1% (\$95,000)
	Total allocated to project		\$95,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.